Yash Vardhan Singh

Greater Noida, India
+91-9305125460 Yashvardhansingh234679@gmail.com
codolio.com/profile/yash_singh
github.com/Yashsingh0607

Education

GL Bajaj Institute of Technology and Management Expected Graduation: 2028
B.Tech in Information Technology *Greater Noida, India*

Relevant Coursework

- Data Structures
- Object-Oriented Programming
- Operating Systems
- DBMS
- Computer Networks
- Cryptography and Network Security

Experience

– Actively seeking internship opportunities and building hands-on projects in software development and cryptography.

Projects

Secure File Encrypter | *Java, Cryptography* March 2026

– Developed a CLI tool to securely encrypt and decrypt files using **AES (CBC mode)**.
– Implemented **PBKDF2-based key derivation (65,536+ iterations)** with salt and IV.
– Designed secure file structure using **Salt + IV + Ciphertext**.
– Implemented error handling for incorrect passwords and invalid inputs.
– Built CLI commands (`encrypt|, decrypt|, --help|).`
– Created installation script for global execution.

Cryptography Implementations | *Java* March 2026

– Implemented **Caesar Cipher** and **Substitution Cipher**.
– Applied modular arithmetic for encryption and decryption.
– Strengthened understanding of cryptographic fundamentals.

Data Structures & Algorithms Practice | *Java*

– Solved 500+ DSA problems in Java..
– Focused on optimized solutions and time complexity analysis.
– Covered arrays, strings, recursion, searching, and sorting techniques.

Technical Skills

Languages: Java, Python (Basic)
Core Concepts: Data Structures & Algorithms, Object-Oriented Programming
Technologies: Cryptography Fundamentals, Modular Arithmetic
Tools: Git, Linux Terminal

Leadership / Extracurricular

- Led a team in Smart India Hackathon (SIH), coordinating development and problem-solving efforts.
- Actively practicing problem-solving and competitive programming.
- Consistent contributor on GitHub with project-based learning.